S.D. COLLEGE OF ENGINEERING & TECHNOLOGY MUZAFFARNAGAR 1997

विद्ययामृतमश्नुते

S.D. COLLEGE OF ENGINEERING & TECHNOLOGY

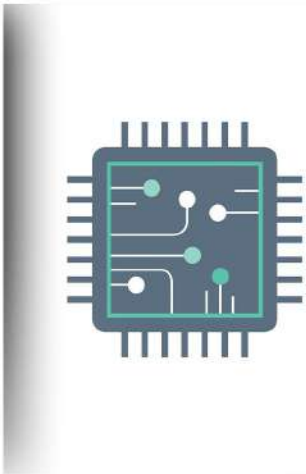
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ECE DEPARTMENT

ELECTROVISION NEWS LETTER

INNOVATION IN ECE 2025







Issue: 01
Volume 01
2025

VISON & MISSION

Vision of Institute

To develop students who can sustain the rigorous industrial requirements in terms of technical skills and motivation and to establish strong linkages with industries and other research organizations of national and international repute.

Mission of Institute

- To establish the institute as an institute of excellence.
- To achieve excellence in imparting technical and professional education.
- To strive to inculcate a sense of integrity and high sense of responsibility amongst students.
- To produce technically skilled manpower who can contribute towards the growth of region as well as the nation.

Vision of Department

To empower students with globally competitive skills in Electronics and Communication Engineering through cutting-edge technology, interdisciplinary learning, and industry collaboration leading towell-qualified engineers who are innovative to their profession and stay aligned with evolving technological advancements.

Mission of Department

The mission of the Department of Electronics & Communication Engineering is equip, guide, and inspire students to excel professionally and grow continuously in their engineering careers. This is accomplished by:

- Providing a conductive learning environment and modern facilities that ensure a strong foundation in engineering principles, preparing students for diverse career paths.
- Encouraging academic excellence and innovative research to enhance innovative creativity, which strengthens the major's regional, national, and international reputation.

FROM THE EDITOR'S DESK

Dear Readers,
The Department of Electronics and Communication Engineering, S.D. College of Engineering & Technology, Muzaffarnagar, is delighted to present to you the **2025 Autumn Edition of ELECTROVISION** – the Department Newsletter.

In this era of rapid technological advancement, innovation in electronics and communication has become the cornerstone of modern development. From Artificial Intelligence to the Internet of Things, from 5G communication to robotics and embedded systems, technology continues to reshape our world. As engineers of the future, it is our responsibility to adapt, explore, and contribute to these evolving domains.

The editorial team of ELECTROVISION is proud to provide a platform that reflects the essence of learning, innovation, and creativity. This newsletter serves as a bridge connecting students, faculty, and readers to the latest trends, achievements, and emerging research in the world of Electronics and Communication.

Technological progress has brought unparalleled convenience and connectivity to human life. It has empowered education, healthcare, industries, and everyday living through the integration of smart and sustainable technologies. At SDCET, we encourage our students to explore these frontiers and turn ideas into impactful innovations.

ELECTROVISION not only highlights academic and technical excellence but also celebrates the diverse activities, accomplishments, and initiatives of our department. We hope this edition inspires our readers to stay curious, stay creative, and continue exploring the ever-expanding universe of technology.

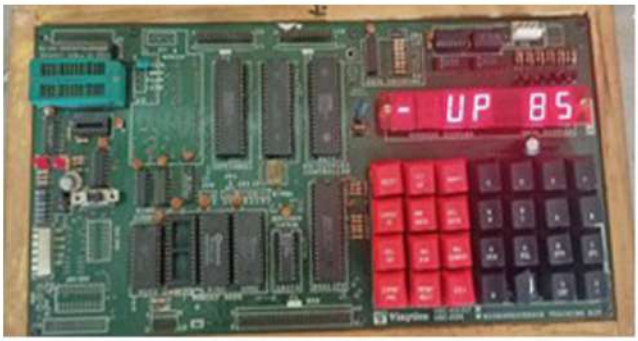
Warm regards,
Editorial Team
Department of Electronics & Communication Engineering
S.D. College of Engineering & Technology, Muzaffarnagar

ABOUT THE DEPARTMENT

The Department of Electronics and Communication Engineering at S.D. College of Engineering & Technology, Muzaffarnagar, was established in the year 1997 with the vision of imparting quality education and fostering innovation in the field of modern electronics and communication systems. The department offers B.Tech and M.Tech programs, focusing on areas such as VLSI Design, Communication Systems, Embedded Technologies, and Microprocessors..

With a strong foundation in innovation and technology, the Department of ECE continues to nurture the engineers of tomorrow, preparing them to excel in the rapidly evolving world of electronics, communication, and digital systems.

LABS







LABORATORIES

- MATLAB [for designing and programming]
- IoT enabled Project & Research Lab
- Communication Engineering Lab
- Digital System Design Lab
- Microprocessor & Microcontroller Lab
- Integrated Circuits Lab
- Network Analysis Lab
- Simulation Lab
- PCB Designing Lab
- Microwave & Optical Communication Lab
- Control System Lab
- Analog Circuits Lab
- Digital Signal Processing Lab

RESEARCH ARTICLES

SURVEY ON WIRELESS INFORMATION ENERGY TRANSFER (WIET) AND RELATED APPLICATIONS IN 6G INTERNET OF NANOTHINGS (IONT)

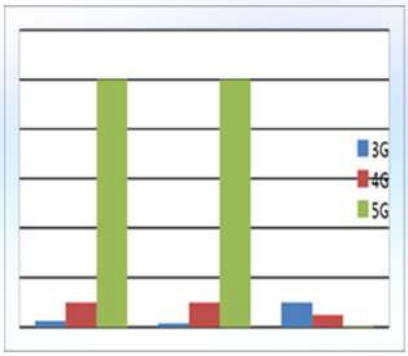
1. INTRODUCTION

In the near future, the 6G technology is expected to integrate millions of devices with the internet due to seamless network capabilities. New dimensions will be explored for small, medium and large scale industries. In this entire portfolio, the new sector which is highly increasing is to minimize the dependency on wired devices for charging of any internet operated device. This is the need of wireless information and energy transfer (WIET) which uses the same electromagnetic wave for serving the communication purpose. Upcoming technologies in the field of wireless communication are expected to utilize the potential of Internet-of-Nanothings (IoNT) vision where many nano-meter wavelength devices are able to communicate and exchange information wirelessly. WIET allows to power up or charge the devices without the need of any wiring infrastructure.

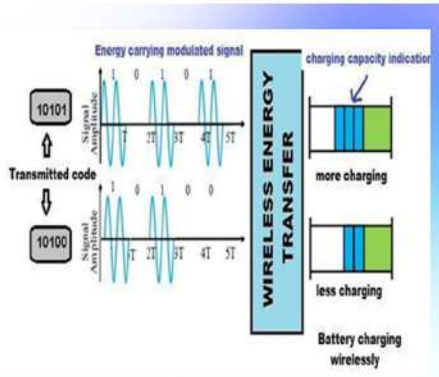
Considering the mechanism used, WIET could be based on three classes viz., magnetic resonant coupling, inductive coupling or electromagnetic radiation [1]. The electromagnetic radiation, also known as radio frequency, enables WIET, and the radio frequency enabled WIET is able to cover long distances at low cost and is able to provide signal to the low power nano-devices. A low power battery operated device can be charged efficiently using the wireless energy without the need of a physical charging unit. The connectivity between the receiver and the transmitter can be divided between the following two modes [1]:

- Mode-1 is the energy flow mode in which the wireless energy is harvested from transmitter to receiver. This energy harvesting (EH) is considered as a key component in upcoming generation IoNT devices since it provides 1) wireless remote charging which minimizes the maintenance and servicing needs of the internetoperated devices, and 2) significant enhancement in energy-efficiency.

- Mode-2 is the information flow mode in which the information contents are transmitted or received through uplink or downlink communication, respectively. Multiple Input Multiple Output (MIMO) technology can be deployed to speed up the efficiency of mode 1 and mode 2 by the activation of multiple antennas at the transceiver which are used to enhance the data rate. It also maximizes the system efficiency, reduces the noise, and minimizes the power consumption so as to achieve the overall goal.



Comparison of bandwidth data rate and latency for 3G to 5G technology.



Various energy carrying code words for WIET.v

As the radio frequency lies in the THz band for 6G communication, more focus should be given for health issues also to minimize the radiation aspect in humans. Hence, in the article, we also suggest future research areas which need research activities for successful implementation of WIET in 6G IoNT as there may be some interference effects, fading, rectenna designing, orthogonal modulation etc.

Dr. Pragati Sharma
Professor, ECE Department

RESEARCH ARTICLES

SEMICONDUCTORS: THE BACKBONE OF MODERN TECHNOLOGY

Semiconductors are the essential building blocks of modern electronics, acting as the hidden brains that make devices work. Semiconductors are materials whose electrical conductivity lies between that of conductors and insulators. They can behave as either, depending on the conditions, making it ideal for electronic devices. This unique property enables the use of semiconductors in creating tiny electronic chips that control the operation of modern devices – not just everyday devices, but also satellites and defense systems. In the Chandrayaan 3 Mission, Vikram lander used Indian made technology and AI to find a safe landing spot all on its own, all by taking complex decisions. Thus, acting as a brain, semiconductor chips help machines make decisions while processing heavy data, giving them iterative direction.

These chips are therefore the backbone of modern electronics, enabling the functioning of devices such as smartphones, computers, electric vehicles, satellites, and even defense systems like Aakshteer.
The chips are made from semiconductor material and can store, process, and transfer information, helping devices perform tasks such as calling, storing data, or sensing electrical signals. Each chip contains millions (or even billions) of microscale switches called transistors, which control electrical signals much like brain cells pass messages in our bodies. It also has other tiny components like resistors, capacitors, and wiring. Together, these elements process and move information.

Semiconductors are at the heart of modern technology. They power essential systems in healthcare, transport, communication, defense, and space. As the world moves toward greater digitalization and automation, semiconductors have become integral to economic security and strategic independence. We can recall the acute shortage of chips that the world faced following the COVID-19 pandemic and the Ukraine-Russia war that affected the manufacturing of cars, mobile phones, and other electronics. The main drivers of growth in the semiconductor industry include the accelerating pace of digitization across sectors, which is increasing the demand for faster, more efficient, and compact electronic components. There is a growing reliance on advanced semiconductor-based systems to process and store vast amounts of data generated by digital platforms, smart devices, and connected infrastructure. The rapid adoption of artificial intelligence and machine learning, both at the edge and in cloud data canters, is fuelling the need for high-performance, energy-efficient chips that can handle complex computing tasks in real time.

Currently, countries like Taiwan, South Korea, Japan, China, and the US dominate the semiconductor industry. Taiwan produces more than 60% of the world's semiconductors, including nearly 90% of the most advanced ones. Such dependence on a single region has exposed global supply chains to significant risks—from pandemics and natural disasters to geopolitical tensions. Recognizing this challenge, many countries are now building secure and diversified supply chains.

Er. Parul Gupta
Asso. Professor, ECE Department

RESEARCH ARTICLES

WHEN THOUGHTS BECOME COMMANDS: THE SCIENCE OF BRAIN-COMPUTER INTERFACES

Consider being able to control a computer, robot, or even a prosthetic limb with only your thoughts. No keyboards, no screens, not even speaking a word—just intent. This is no longer fantasy; it is the fast-evolving world of brain-computer interfaces (BCIs), an area at the convergence of neuroscience, engineering, and artificial intelligence.

A BCI at its most basic is a system that converts neural activity into useful commands for an external device. The brain talks to itself in electrical terms through the firing of neurons, which produce action potentials. BCIs access these through various means. Invasive BCIs employ small lectrodes implanted into the brain's cortex, which offer high-resolution readings. Semi-invasive BCIs rest under the skull oragainst the surface of the brain, minimizing surgical risk but providing adequate signal quality. Non-invasive BCIs employ scalp-borne sensors like EEG (electroencephalography) or fNIRS (functional near-infrared spectroscopy) to pick up brain waves, although these are less accurate.

The function of a BCI is as follows:
1. Signal Acquisition – The sensors or electrodes record electrical signals or hemodynamic fluctuation in the brain.
2. Signal Processing – Signals are filtered to eliminate noise due to muscle movements, eye movements, or environmental sources.
3. Feature Extraction – Critical aspects of the signal, such as frequency patterns or spike timing, are extracted to describe the user's intended action.
4. Decoding / Classification – Sophisticated algorithms, frequently underpinned by machine learning, transform these features into precise commands, such as moving a cursor or operating a robotic arm.
5. Adaptation and Feedback – The system gives immediate feedback to the user, enabling both the machine and the individual to become more precise with time. BCIs find its applications mostly in medical field, enabling people with paralysis, spinal cord injury, or neurodegenerative diseases to regain function. For instance, a patient can operate a robotic arm or even talk through a text-to-speech system using neural activity alone. New research is investigating decoding speech and language directly from brain signals, which could revive communication for speech-impaired individuals. Other potential uses are neuro rehabilitation, therapy for mental illness through targeted brain stimulation, and even futuristic human-computer interaction during gaming or virtual reality.
Recent breakthroughs demonstrate the potential of BCIs on multiple fronts. Companies like Neuralink are developing high-density implants capable of recording thousands of neurons simultaneously, while AI-driven algorithms are decoding more complex intentions faster and more accurately than ever. Brain-computer interfaces are not only future technology—they are a revolutionary means of linking the human brain to machines. By recording the language of neurons, translating thought into motion, and providing individuals with new skills, BCIs are the future of human-machine interaction. As technology and science evolve, the boundary between idea and action might disappear soon, revealing possibilities only seen in sci-fi.

Sundus Khan
ECE, 2nd year

RESEARCH ARTICLES

WIRELESS POWER TRANSFER: TECHNOLOGIES, APPLICATIONS, AND FUTURE PROSPECTS

Abstract
Wireless Power Transfer (WPT) has emerged as a revolutionary technology enabling the transmission of electrical energy without physical connectors. This research paper provides a comprehensive overview of WPT technologies,including inductive coupling, resonant inductive coupling, capacitive coupling, and radio frequency (RF) energy harvesting. The paper explores fundamental principles, circuit designs, practical applications ranging from consumer electronics to electric vehicles, and discusses efficiency challenges and future research directions. Key findings indicate that resonant inductive coupling offers optimal efficiency for mid-range power transfer, while emerging technologies promise extended range capabilities for future applications.

1.1 Background

The concept of wireless power transfer dates back to the pioneering work of Nikola Tesla in the late 19th century. Tesla envisioned a world where electrical energy could be transmitted through the air, eliminating the need for extensive wire networks. While his ambitious projects like the Wardenclyffe Tower were not commercially successful during his lifetime, they laid the groundwork for modern WPT technologies.

2. Wireless Power Transfer Technologies

2.1 Inductive Power Transfer (IPT)

Inductive Power Transfer operates at close range (typically 1-10 mm) and is the technology behind Qi wireless charging
standard used in smartphones. The system consists of a primary coil driven by an AC source (typically 100-200 kHz) and a
secondary coil that receives the induced power.

2.1.1 Key Components

High-Frequency Inverter: Converts DC to AC at operating frequency (85-200 kHz)
Resonant Capacitors: Tune both primary and secondary circuits to resonance
Transmitter Coil: Creates alternating magnetic field
Receiver Coil: Captures magnetic flux and converts to electrical energy
Rectifier: Converts AC back to DC for device charging
Control Circuit: Manages power flow and communication

2.2 Resonant Inductive Coupling

Resonant inductive coupling extends the range of IPT by operating both transmitter and receiver circuits at their resonant frequency. When the transmitting and receiving coils are tuned to the same resonant frequency, power transfer efficiency significantly increases, enabling transfer over distances of several centimeters to meters. The resonant frequency is determined by: $f_0 = 1 / (2\pi\sqrt{(LC)})$ Common resonant frequencies for WPT systems range from 85 kHz (Qi standard) to 6.78 MHz (AirFuel Alliance).

2.3 Capacitive Power Transfer

Capacitive Power Transfer (CPT) uses electric fields instead of magnetic fields. The system employs metal plates that form capacitors, with the air gap or dielectric material between them serving as the capacitor's dielectric. CPT is particularly suitable for applications requiring power transfer through metal barriers.

3. Conclusion The convergence of WPT with IoT, artificial intelligence, and advanced materials will enable autonomous charging systems,self-powered sensor networks, and ubiquitous energy availability. As manufacturing costs decrease through economies of scale and standardization efforts mature, WPT will become invisible infrastructure supporting our increasingly wireless world. Future developments in dynamic charging, long-range power transfer, and space-based applications will further expand WPT's impact.

Piyush Gaur
ECE, 3rd Year

RESEARCH ARTICLES

QUANTUM ERROR CORRECTION

Quantum computers are often described as the future of technology. They promise to solve problems that are impossible for today's fastest supercomputers. However, one of the biggest challenges for quantum computing is quantum error correction (QEC).The main issue is that the information stored in a quantum computer is easily lost due to noise and environmental disturbances. To make quantum computers practical, scientists need powerful ways to detect and correct these errors.

Like normal computers, quantum computers use their own version of bits — called quantum bits or qubits. These qubits are different from classical bits because they can exist in a superposition state of both 0 and 1 simultaneously. This unique property makes quantum computers exponentially faster than traditional ones. In classical computing, error correction is well-developed. The Low-Density Parity Check (LDPC) code is widely used to ensure reliable data transmission. These codes work efficiently and can reach the maximum possible efficiency, known as the channel capacity.

Unlike classical systems, quantum computation relies on entanglement, which allows quantum information to be transmitted in ways that appear faster than the classical speed of communication. For decades, researchers have tried to find effective solutions for Quantum Error Correction (QEC). The ultimate goal has been to reach the hashing bound — a theoretical limit that defines how much quantum information can be sent reliably through a noisy channel. Until recently, no known quantum code had come close to achieving this limit.

A recent study by Daiki Komoto and Kasai has changed that. These scientists developed a special quantum LDPC code that approaches the hashing bound. This code helps decode errors and track stored information without requiring enormous computational power. They used an advanced version of the LDPC code combined with a clever decoding method known as the Product-Sum Algorithm. This algorithm can handle both major types of quantum errors — bit-flip and phase-flip — simultaneously. It also addresses a problem called the error floor, where error correction stops improving after a certain point.

When tested through simulations, the new code showed excellent performance. It worked reliably even when the error rate was very small, and its performance closely matched the theoretical hashing bound. More importantly, the decoding process scaled efficiently as the number of qubits increased — a key factor for building scalable quantum computers in the future. This breakthrough brings us much closer to developing fault-tolerant quantum machines that can run for long durations without being disturbed by noise. Although further refinements are needed — especially in handling specific types of quantum errors — this research marks a major step forward in the journey toward practical quantum computing.

Mayank Sharma
ECE, 2nd Year

DID YOU KNOW??

- The term "Wi-Fi" doesn't actually mean "Wireless Fidelity" — it's just a catchy brand name created for marketing!
- A single modern micro-SD card can hold more data than an entire room of computers from the 1950s.
- The first integrated circuit (IC) was invented in 1958 by Jack Kilby, who later won the Nobel Prize for it.
- The camera used on the Voyager spacecraft in 1977 had less resolution than most mobile phone cameras today!
- The "G" in 4G and 5G stands for Generation — each generation marks a major leap in speed and connectivity.
- The world's first transistor was made at Bell Labs in 1947 — it was so big it could barely fit in your hand. Today, billions of them fit inside a single microchip!
- Modern AI chips like Google's TPU and NVIDIA's GPU can perform trillions of calculations per second, powering everything from chat bots to self-driving cars!
- IoT (Internet of Things) devices now outnumber humans on Earth — with over 15 billion connected sensors and gadgets worldwide.
- A single Apple M1 chip contains 16 billion transistors — that's more than the total population of Earth, packed inside a few centimeters!
- The size of a transistor in today's chips is as small as 2 nanometers — that's about 50,000 times thinner than a human hair!
- During chip fabrication, even a tiny dust particle can ruin an entire silicon wafer — that's why chip plants operate in ultra-clean rooms 10,000 times cleaner than a hospital!
- Photo-lithography, the process used to create chip patterns, uses UV light to etch nanoscale circuits — newer methods now use Extreme Ultraviolet (EUV) light for even smaller designs.
- The human ear can detect frequencies between 20 Hz and 20,000 Hz, while ultrasonic sensors used in electronics can detect sounds up to 40,000 Hz or more!

QUOTES

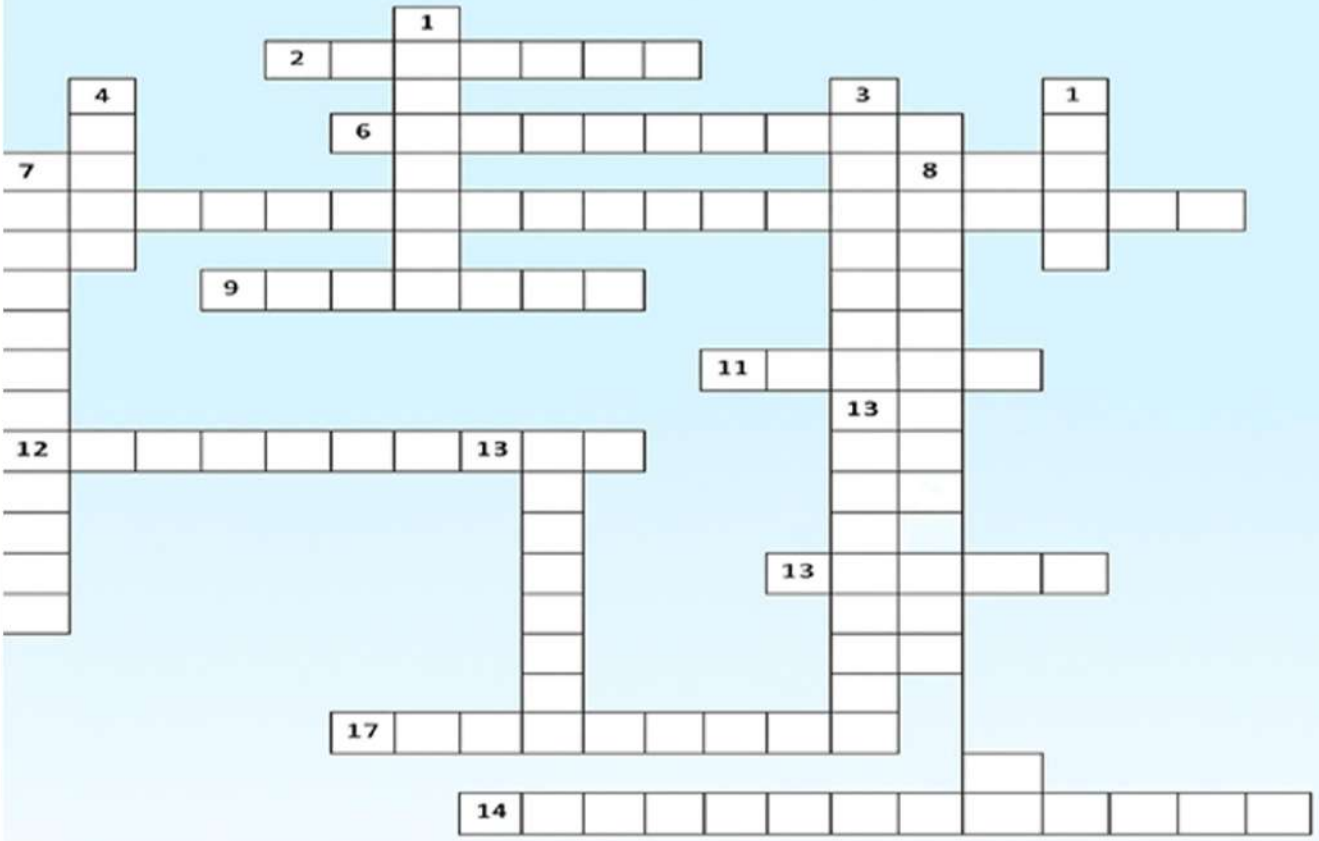
"Every small step forward is a victory over doubt." – Eleanor Grant

"Failures are not roadblocks;; they are road signs guiding success." – A. M. Carter

"Arise, awake, and stop not till the goal is reached." – Swami Vivekananda

"You cannot change your future, but you can change your habits, and surely your habits will change your future." – **Dr. A.P.J. Abdul Kalam.**

Crossword



Across

2. Smaller particles within an atom
4. What is the name of the global comuter network?
5. Intel is thelargest manufacturer of what type of component?
8. System for electronic conmmunication
11. 1,000,000,000 = 1__
13. His code for decoding text
16. First computer made by Apple Inc.
17. Wi-Fi stands for ____ fidelity

Down

1. Device for detecting electric current
3. Snapdragon, Cortex and Kirin are examples of ____
6. Devices connected to a network
7. Number of bits in one byte
9. Morse uses dots and dashes in Its code
10. Usually used to generate electrical power
12. Digital Information is represented by 1s and 0s

DEPARTMENTAL EVENTS

"Every small step forward is a victory over doubt." – Eleanor Grant

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Muzaffarnagar
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Prof. Basheshwar Prasad
Dept. of Chemical Engineering,
IIT Roorkee



Prof. Vimal Chandra Srivastava
Dept. of Chemical Engineering,
IIT Roorkee





IIT, Roorkee Professor Vimal Chandra Srivastava giving GATE 2025 details

ALUMNI INTERACTION

Ms. Parul , 2002 batch passout shared her thoughts with students [08.07.2025]





Students maintain regular interaction with alumni, which provides them with valuable industry exposure and practical insights. Through seminars, mentoring sessions, and experience-sharing interactions,

Alumni guide and motivate students to align their academic learning with real-world professional requirements. These engagements not only inspire students but also help them build strong networks and prepare for successful careers.

CLUBS & ACTIVITIES

- **IIRS-ISRO** online course: **"GLOBAL NAVIGATION SATELLITE SYSTEM"** successfully completed.
- Technical Club Events for students over all development









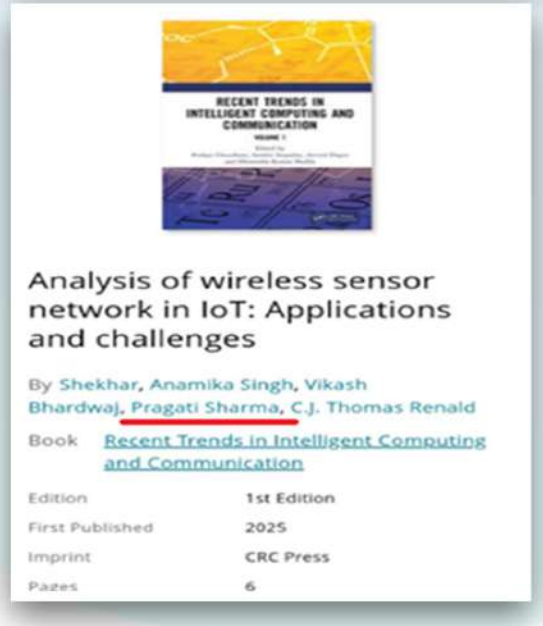
Students actively engaging in hands-on learning and simulation to gain practical exposure and technical skills.

Students regularly participate in interactive sessions, laboratory demonstrations, and field visits in Muzaffarnagar, Uttar Pradesh. These activities aim to bridge the gap between theoretical knowledge and real-world applications.

FACULTY CORNER









Dr. Pragati Sharma has been honored with the Education Excellence Award 2025 by I2OR for her remarkable contributions in academics and research. She was honored with the Certificate of Reviewer Recognition at the 3rd IEEE International Conference on Networks, Multimedia, and Information Technology (NMITCON-2025), held at NMIT, Bangalore. She also coordinated in the Virtual Labs Training Program organized by the Department of Electrical Engineering, IIT Roorkee, under the National Mission on Education through ICT (NMEICT).

Alongside, **Ms. Parul Gupta** has also achieved certification from **MSME Technology Centre**, showcasing the institution's strong focus on excellence, innovation, and professional growth.

STUDENTS CORNER



अंतरिक्ष अकादमी में ट्रेनिंग के लिए हुआ चयन

संवाद न्यूज एजेंसी

मुजफ्फरनगर। जानसठ रोड स्थित एसडी कॉलेज ऑफ इंजीनियरिंग के दो छात्र वासु और अमर रिमोट सेंसिंग और जीआईएस का कौशल प्रशिक्षण प्राप्त करेंगे। इलेक्ट्रॉनिक्स एंड कम्युनिकेशन इंजीनियरिंग के दोनों छात्रों का चयन प्रीप्लेसमेंट कौशल प्रशिक्षण कार्यक्रम के तहत हुआ है। इस

प्रशिक्षण के दौरान वह अंतरिक्षयान से जुड़ी बारीकियों को सीखेंगे। कॉलेज के निदेशक सिद्धार्थ शर्मा ने बताया कि दोनों छात्र बीटेक तृतीय वर्ष के छात्र हैं। रिमोट सेंसिंग एवं जीआईएस के प्रशिक्षण के लिए उनका चयन लिखित और साक्षात्कार प्रक्रिया के जरिए हुआ। दोनों छात्र कॉलेज के सर्किट लर्निंग कार्यक्रम में

एस डी कॉलेज ऑफ इंजीनियरिंग के छात्र वासु और अमर। फोटो : स्वयं

काफी सक्रिय रहे। उनकी रुचि और ज्ञान के बलबूते उन्होंने कठिन परीक्षा को पार कर अपना स्थान बनाया।

कॉलेज के सचिव अनुभव कुमार ने छात्रों को शुभकामनाएं दीं। उन्होंने कहा कि राष्ट्रीय स्तर प्रशिक्षण कार्यक्रम स्थान बनाना गौरव की बात है। भारत अंतरिक्ष अकादमी द्वारा संचालित यह विशेष प्रशिक्षण कार्यक्रम छात्रों को अंतरिक्ष प्रौद्योगिकी के क्षेत्र में आधुनिक तकनीकों से अवगत कराएगा। उनके कौशल को निखारने में सहायक होगा।



एस.डी. कॉलेज ऑफ इंजी. एंड टेक्नोलॉजी के छात्रों ने रचा नया कीर्तिमान

– सनी कुमार और अमन चंदेल का डीआरडीओ में चयन

मुजफ्फरनगर, 01 अगस्त (खास)। जानसठ रोड स्थित एसडी कॉलेज ऑफ इंजीनियरिंग एंड टेक्नोलॉजी में मुजफ्फरनगर के इलेक्ट्रॉनिक्स एवं कम्युनिकेशन ब्रांच के छात्रों ने एक नई उपलब्धि हासिल की है। कॉलेज के होनहार छात्र सनी कुमार और अमन चंदेल का चयन देश के प्रतिष्ठित रक्षा अनुसंधान एवं विकास संगठन (डीआरडीओ) की इकाई टिरेम इलेक्ट्रॉनिक्स एप्लीकेशन लैबोरेटरी देहरादून में पेड इंटर्नशिप-2025 के लिए हुआ है। इस वर्ष पूरे देश से कुल 14 छात्रों का चयन हुआ। एस. डी. कॉलेज ऑफ इंजीनियरिंग एंड टेक्नोलॉजी में मुजफ्फरनगर के ही कुमार और अमन चंदेल भी शामिल हैं। यह इंटर्नशिप 11 अगस्त 2025 से प्रारंभ होगी, जिसमें चयनित छात्रों को प्रयोगशाला में अत्याधुनिक रक्षा इलेक्ट्रॉनिक्स परियोजनाओं पर कार्य करने का अवसर मिलेगा। सचिव अनुभव कुमार ने इस उपलब्धि को संस्थान के लिए गौरवपूर्ण बताते हुए कहा कि हमारे छात्रों का डीआरडीओ जैसी प्रतिष्ठित संस्था में चयन होना हमारे शिक्षकों की कड़ी मेहनत, सुचारु शिक्षा और छात्रों की लगन का परिणाम है। हमें विश्वास है कि सनी और अमन इस अवसर का पूरा उठाकर कॉलेज और जिले का नाम और रोशन करेंगे। निदेशक डॉ. शर्मा ने कहा कि यह हमारे छात्रों की योग्यता और तकनीकी क्षमता को दर्शाता है। डीआरडीओ में कार्य करना किसी भी छात्र के लिए एक सपना होता है, और हमें गर्व है कि हमारे विद्यार्थी इस मुकाम तक पहुंचे हैं। हम उन्हें इस इंटर्नशिप के लिए शुभकामनाएं देते हैं और आशा करते हैं कि वे आने वाले समय में देश की तकनीकी प्रगति में महत्वपूर्ण योगदान देंगे। कॉलेज के इलेक्ट्रॉनिक्स एवं कम्युनिकेशन विभाग के प्राध्यापकों ने भी इस उपलब्धि पर गर्व जताया और कहा कि यह सफलता पूरे विभाग के लिए प्रेरणास्रोत है। उन्होंने छात्रों को बधाई देते हुए भविष्य में और भी ऊंचाइयों पर पहुंचने की शुभकामनाएं दीं।

सीओ सदर एवं थाना प्रभारी ने ग्रामवासियों के साथ एम.जी. पब्लिक स्कू



ECE students **Vashu Kumar** and **Sunny Kumar** have been selected for training at the **Indian Space Research Academy**. Their achievement has brought pride to the institution.

Further Sunny Kumar and Aman Chandel have brought laurels by being selected for training at the prestigious **Defence Research and Development Organisation (DRDO)**. Their achievement highlights the institution's commitment to nurturing talent and encouraging innovation in science and technology.

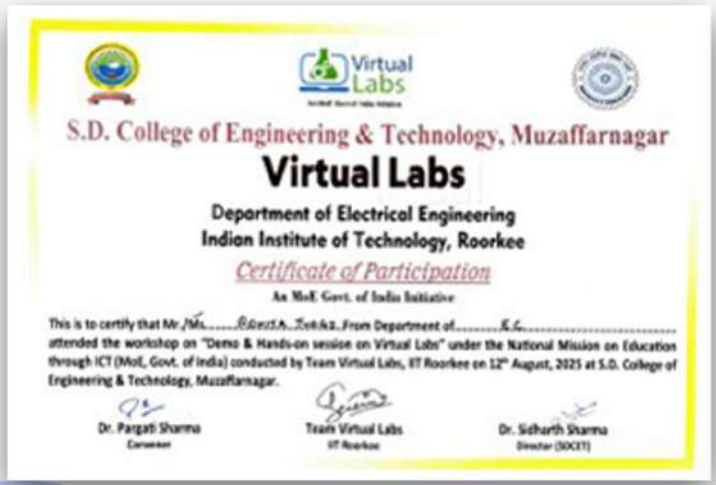
STUDENTS CORNER


Dr. A.P.J. ABDUL KALAM TECHNICAL UNIVERSITY, UTTAR PRADESH, LUCKNOW
DR. ABDUL KALAM INTER-TECHNICAL UNIVERSITY
LITERARY, MANAGEMENT AND TECHNICAL FEST, 2024-25
(ZONAL LEVEL)
Organized at
S. D. COLLEGE OF ENGINEERING & TECHNOLOGY, MUZAFFARNAGAR
(Zonal Centre)
CERTIFICATE
This is to certify that Mr./Ms. NIRVANSH PAL Student of S.D.C.E.T. MUN
Roll No. 2301210307 has participated in Ro Bo RACE Event and Secured ..... position at Dr. Abdul Kalam Inter-Technical University Literary Management and Technical Fest, 2024-25 (Zonal Level) held on 27th-28th November, 2024.
Manoj Kumar Jha — Dr. Sidharth Sharma — Prof. O. P. Singh — Prof. J.P. Pandey


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Professor Level (AICTE Institute), Coordinator
East Point College of Engineering and Technology
Dr. Sunil Luthra
Director & Bureau Head
Training and Learning Bureau, AICTE


S.D. COLLEGE
OF ENGINEERING & TECHNOLOGY
Jansath Road, Muzaffarnagar, (U.P)- 251001
CERTIFICATE
of Appreciation
This is to certify that Mr./Ms. Nirvansh Pal
Son/Daughter of Mr. Manish Pal has been awarded this appreciation certificate for his/her Participation in Internal Hackathon
We wish you good luck for future!
"A winner is someone who sets their goals, commits to them wholeheartedly, and pursues them with unwavering determination and full effort."
Director
Date 13th/15th July 2024
Secretary


Virtual Labs
S.D. College of Engineering & Technology, Muzaffarnagar
Virtual Labs
Department of Electrical Engineering
Indian Institute of Technology, Roorkee
Certificate of Participation
An MoE Govt. of India Initiative
This is to certify that Mr./Ms. Priya Tyagi From Department of E.E. attended the workshop on "Demo & Hands-on session on Virtual Labs" under the National Mission on Education through ICT (MoE, Govt. of India) conducted by Team Virtual Labs, IIT Roorkee on 12th August, 2025 at S.D. College of Engineering & Technology, Muzaffarnagar.
Dr. Pargati Sharma — Coordinator
Team Virtual Labs — IIT Roorkee
Dr. Sidharth Sharma — Director (SDCET)


INTERNATIONAL YOGA DAY 2024
I PLEDGE TO INCLUDE YOGA AS A LIFESTYLE IN PRACTICING IT REGULARLY, BOTH FOR MYSELF AND WITH MY FAMILY.
Certificate of Pledge
This is to certify that
Priya Upadhyay
from
Dr. A.P.J. Abdul Kalam Technical University, Lucknow
has taken the pledge on 18-Jun-2024
Yoga for Harmony & Peace

STUDENTS CORNER



CERTIFICATE
OF COMPLETION

THIS CERTIFICATE IS PROUDLY PRESENTED TO

VASHU

successfully completed the 40 days hands-on training program on **Drone Technologies and Applications**, at **S. D. College of Engineering (SDCET), Muzzafarnagar**, implemented by **IndoplanetX Space Vault and Research Pvt. Ltd., Roorkee** and allocated **A** grade for their performance during training.

The training program included theoretical sessions, practical workshops, and project-based learning, equipping the participant with essential knowledge and skills in drone operations, maintenance, aerodynamics, safety protocols, and real-world applications of drone technology.

Ankit Kumar
FOUNDER | CEO

Arun Kumar
DIRECTOR



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20250411865673

Issue Date
11 April, 2025

Shalini Tewari
Chief Operating Officer



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Sadiya Tyagi

for participation in the **Career Guidance Session** organized by **"E-Cell IIT Hyderabad"** in association with Gradguru Innovations conducted on 8/30/2025
We congratulate you on this achievement and wish you the best.

ISO 9001:2015 COMPANY

Issued by :

Career counsellor
GradGuru



ISVR/2025/ERN/002

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AMAN CHANDEL

successfully completed the 40 days hands-on training program on **Drone Technologies and Applications**, at **S. D. College of Engineering (SDCET), Muzzafarnagar**, implemented by **IndoplanetX Space Vault and Research Pvt. Ltd., Roorkee** and allocated **B+** grade for their performance during training.

The training program included theoretical sessions, practical workshops, and project-based learning, equipping the participant with essential knowledge and skills in drone operations, maintenance, aerodynamics, safety protocols, and real-world applications of drone technology.

Ankit Kumar
FOUNDER | CEO

Arun Kumar
DIRECTOR



CERTIFICATE
OF COMPLETION

THIS CERTIFICATE IS PROUDLY PRESENTED TO

PRIYA UPADHYAY

successfully completed the 40 days hands-on training program on **Drone Technologies and Applications**, at **S. D. College of Engineering (SDCET), Muzzafarnagar**, implemented by **IndoplanetX Space Vault and Research Pvt. Ltd., Roorkee** and allocated **A** grade for their performance during training.

The training program included theoretical sessions, practical workshops, and project-based learning, equipping the participant with essential knowledge and skills in drone operations, maintenance, aerodynamics, safety protocols, and real-world applications of drone technology.

Ankit Kumar
FOUNDER | CEO

Arun Kumar
DIRECTOR



भारत अंतरिक्ष अकादमी
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दूरभाष संख्या : 011–44749707, 8130317917
ईमेल: info@isa.ac.in, contact@isa.ac.in
वेबसाइट: www.isa.ac.in

India Space Academy
Department of Space Education (ISW)
BA/ 14 B Janakpuri
New Delhi - 110058, India
Telephone no: 011-44749707, 8130317917
Email : info@isa.ac.in, contact@isa.ac.in
Website: www.isa.ac.in

Internship Completion Letter

Admission No.- 176430

To,
AMAN CHANDEL
S.D. COLLEGE OF ENGINEERING & TECHNOLOGY,MUZAFFARNAGAR

Subject: Confirmation of Completion of Internship and Project Report Submission.

Dear Candidate,
This is to formally certify that AMAN CHANDEL has completed a four-week internship with India Space Academy serving as a "Remote Sensing and GIS Intern" during our Summer Training Program held from July 15, 2025, to August 14, 2025. During this period, the candidate actively participated in structured training sessions and project-based work focused on Remote Sensing and Geographic Information Systems (GIS).

The candidate demonstrated strong analytical acumen, coding proficiency, and effective problem-solving abilities while contributing to the assigned project(s). We sincerely appreciate the dedication, diligence, and meaningful contributions exhibited throughout the internship tenure.

We wish you all the best in your future endeavors and career pursuits. We are confident that the skills and experience you gained here will serve you well.

The candidate completed and submitted the report of the following project (s):

- **Mapping Water Bodies using NDWI: The project utilized NDWI derived from multi-spectral satellite imagery to identify, extract, and delineate water features. The analysis enhanced the visibility of rivers, lakes, and reservoir**

Best regards,

निदेशक कार्यालय/Director Office

STUDENTS CORNER













STUDENTS EXPOSURE

- Hackathons [20.09.2025]

- Alumni Talk [08.07.2025 & 08.08.2025]









Students of the **Electronics & Communication Engineering (ECE)** Department have been actively and continuously participating in various **Tech Fests, Hackathons**, and **Workshops/Training** Programs at institutional, regional, and national levels. These platforms not only enhance their technical knowledge but also encourage innovation, teamwork, and problem-solving skills.

Participation in **Workshops and Training Programs** further strengthens their hands-on learning in emerging technologies such as IoT, AI, Robotics, VLSI, and Communication Systems.

Such consistent participation reflects the department's commitment to fostering a culture of **technical excellence, innovation, and continuous learning** among its students.

ECE Department students successfully designed and developed a Bluetooth Speaker as part of their practical learning initiative.

BEYOND ACADEMICS @ECE





''गरबा-ए-ग्लैम-2025'' में छात्र छात्राओं ने खूब खेला डांडिया

मुजफ्फरनगर। जानसठ रोड स्थित एस. डी. कॉलेज ऑफ इंजीनियरिंग एंड टेक्नोलॉजी में नवरात्रि के पावन अवसर पर ''गरबा ए ग्लैम-2025'' का रंगारंग और हर्षोल्लासपूर्ण आयोजन संपन्न हुआ। कार्यक्रम में छात्र-छात्राओं ने पारंपरिक परिधानों में सजधजकर गरबा और डांडिया की मनमोहक प्रस्तुतियां दीं, जिससे पूरे माहौल में उत्सव की चमक और उमंग बढ़ गई।कार्यक्रम के अंतर्गत डांडिया प्रतियोगिता का भी आयोजन किया गया, जिसमें प्रथम पुरस्कार वासुदेव और आरुषि तथा द्वितीय पुरस्कार विशेष और हिमांशी को प्रदान किया गया। विजेताओं को ट्रॉफी और प्रमाण पत्र देकर सम्मानित किया गया। संस्थान के सचिव अनुभव कुमार ने छात्रों को संबोधित करते हुए कहा कि भारतीय संस्कृति और परंपराओं की जड़ें हमें सदैव एकता, भाईचारे और सकारात्मक ऊर्जा की प्रेरणा देती हैं। इस प्रकार के सांस्कृतिक आयोजन विद्यार्थियों के व्यक्तित्व विकास और रचनात्मकता को निखारने में सहायक होते हैं। निदेशक डॉ. सिद्धार्थ शर्मा ने कहा कि सांस्कृतिक कार्यक्रम न केवल मनोरंजन प्रदान करते हैं, बल्कि आत्मविश्वास, सामूहिकता और नेतृत्व



क्षमता के विकास में भी सहायक होते हैं। उन्होंने नवरात्रि पर्व को शक्ति, श्रद्धा और भक्ति का प्रतीक बताते हुए कहा कि छात्रों को परंपराओं के साथ आधुनिकता का संतुलन बनाकर आगे बढ़ना चाहिए। कार्यक्रम की विशेष अतिथि एवं संस्थान की पुरातन छात्रा वैष्णवी भारद्वाज, सीनियर सॉफ्टवेयर इंजीनियर, यूनाइटेड हेल्थ ग्रुप ने कहा कि इस कॉलेज की शिक्षा और मार्गदर्शन ने उनके करियर को नई दिशा दी। उन्होंने छात्रों से प्रेरणा लेते हुए भारतीय संस्कृति और मूल्यों से जुड़े रहने और अपने सपनों को साकार करने का संदेश दिया। पूरे कार्यक्रम ने नवरात्रि के पर्व की भव्यता को और भी शानदार बना दिया और छात्रों में सांस्कृतिक उत्साह और रचनात्मकता को प्रोत्साहित किया।

BEYOND ACADEMICS @ECE

Anirudh Kumar Sharma
Owner, BBSRC India



Praveen Bhardwaj
S.D.O,BSNL, Bulandshahar



Rahul Singh
IES , Indian Railways,Lucknow



Akshat Agarwal
Service Portfolio Director ,
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EDITORIAL TEAM

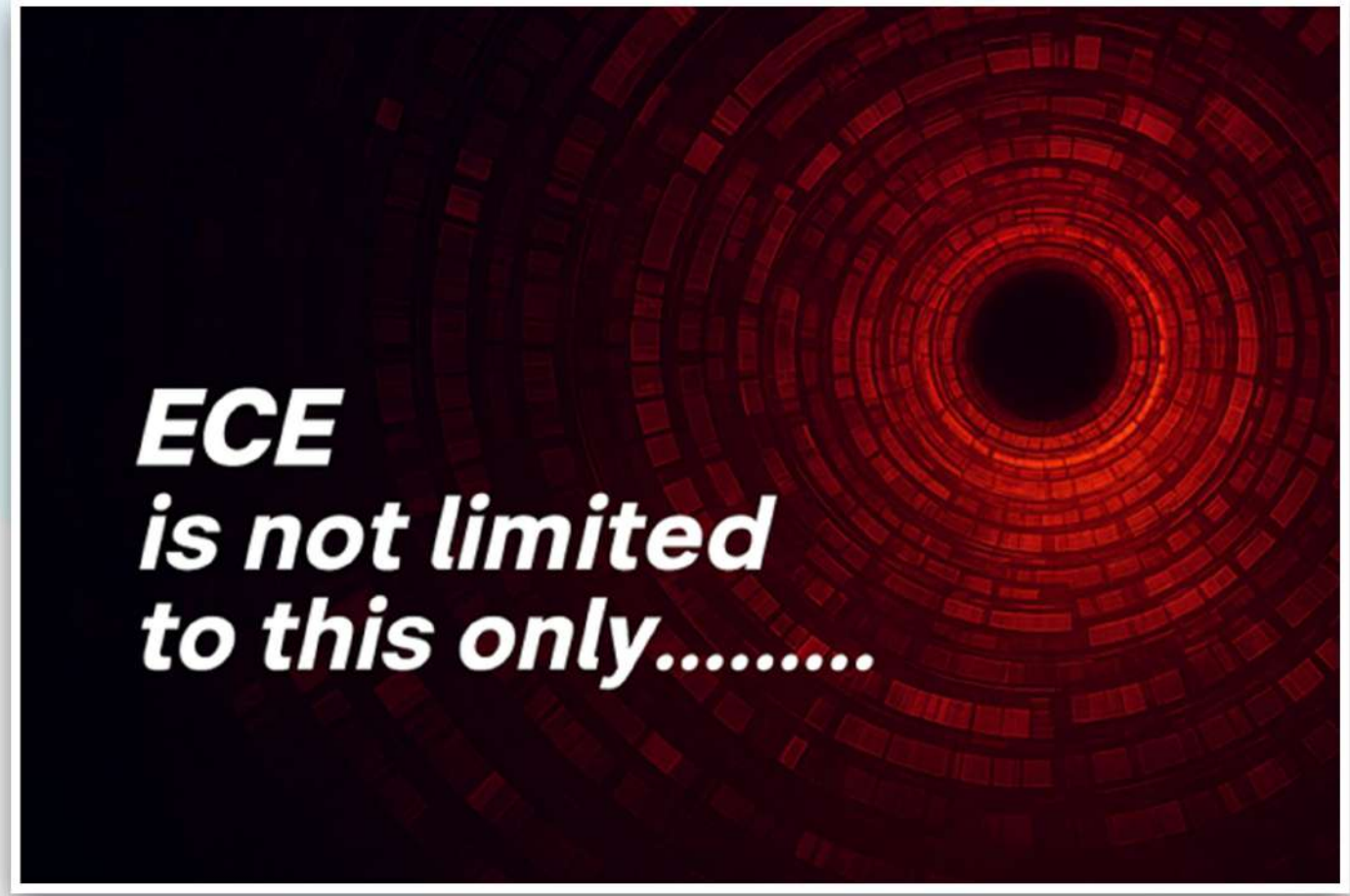
Faculty Editors:

- Dr. Pragati Sharma [9219053412]
- Er. Parul Gupta [9410620018]

Student Editors:

- Vashu/ ECE/ IV Year
- Hani/ ECE/III Year
- Pradeep Gautam/ECE/II Year
- Mayank Sharma/ECE/II Year

Give your suggestions at : 9219053412



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